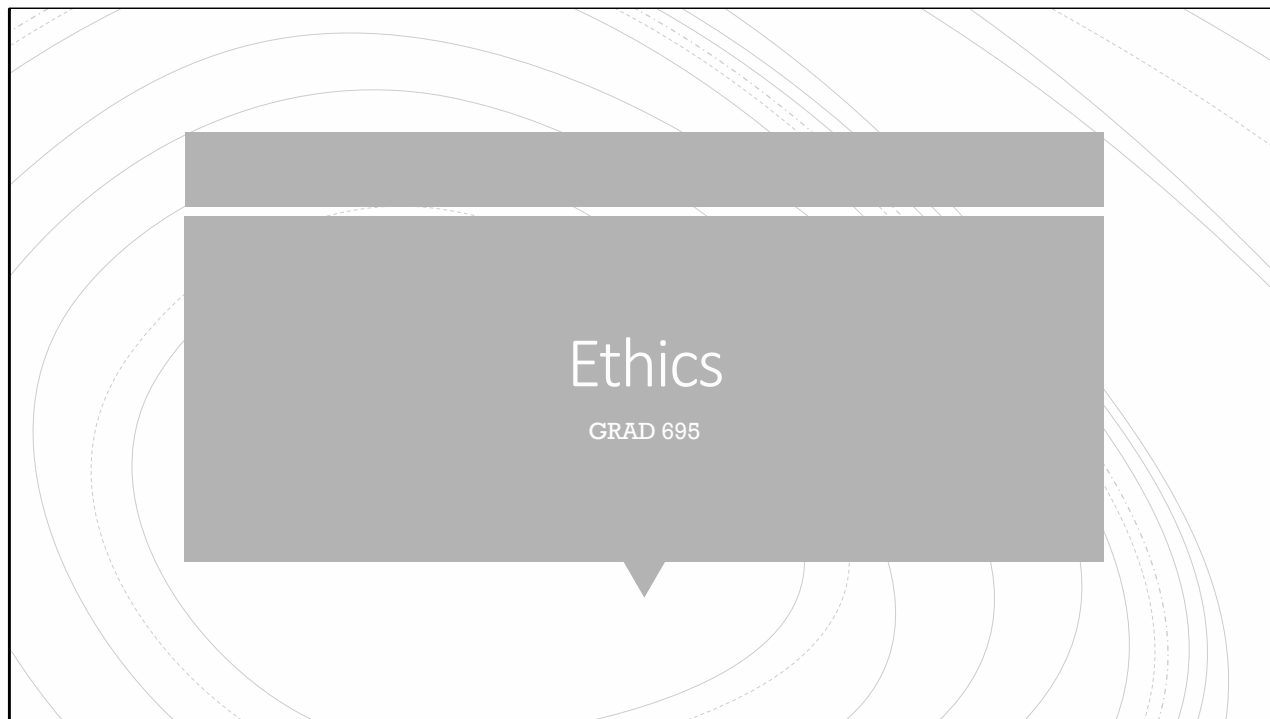
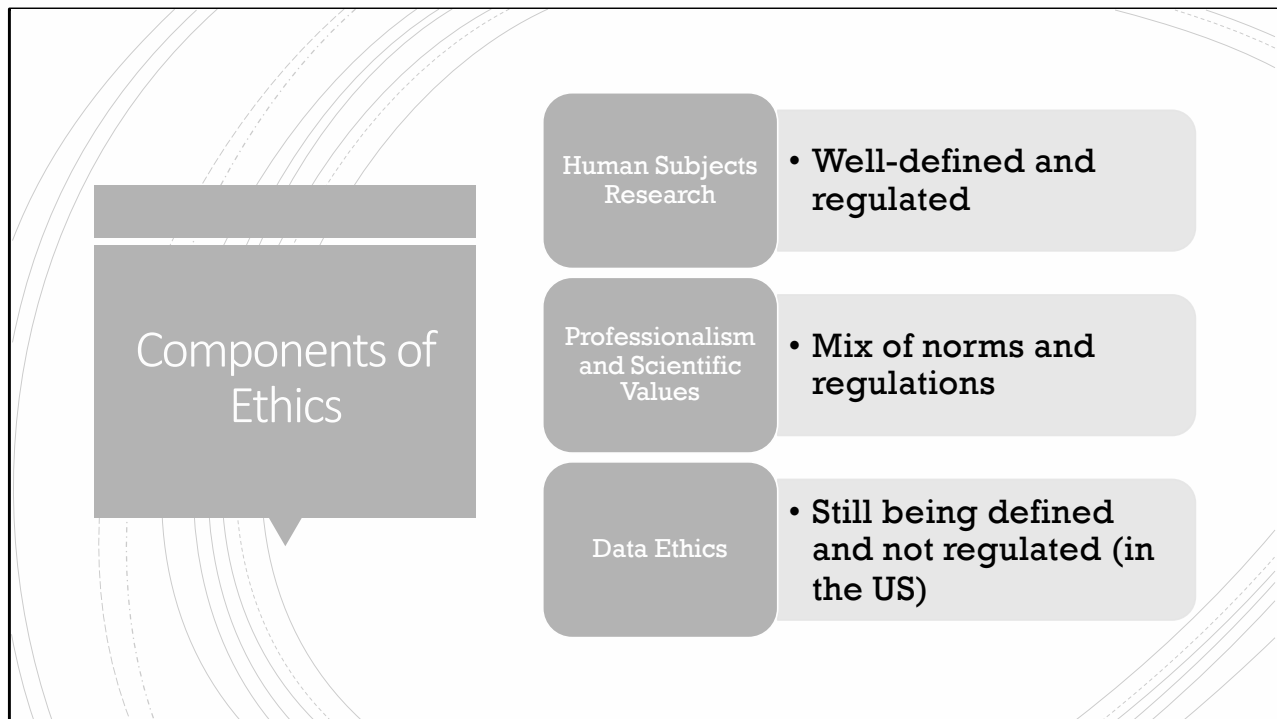




This might be overlapping a little bit if you've taken the ethics course, but we'll just try to hit the highlights here and I'll give you some resources if you want to learn more about anything specific.



So there are several components when it comes to research ethics.

The first is human subjects research (and also animal research) where most of the kind of academic applications of research. This component is well defined and well regulated through things like the IRB and federal regulations. I'll only briefly touch on that tonight as the CITI training and IRB assignments will cover those issues in detail.

A second component of ethics is professionalism and scientific values; there are some laws and regulations around this, but mostly a matter of scientific norms that we practice.

The third component is data, and now AI ethics, which you likely discussed in ANLY 620. This is a constantly evolving set of rules and regulations which is becoming more formalized with laws such as the GDPR and EU AI Act but there are many issues still being debated and discussed.

- Regulated in the US by 45 CFR 46 (1974 – present)
 - Core Principles: Informed Consent, Fairness & Equity, Do No Harm
- At the university level is overseen by Institutional Review Board
 - [HU IRB Sharepoint Hub](#) contains guides, templates, and resources for conducting ethical research
- Assignments below cover research from an institutional viewpoint in more depth; both are required to successfully complete this course:
 - IRB Overview
 - CITI Ethics Training (complete Group 1, 2, or 3 depending on your research interest; Group 2 for Social and Behavioral Research is recommended for Analytics)

Human Subjects Research

For human subjects research, you'll get the kind of gist of this in two of the first assignments, the IRB overview assignment and then the CITI training.

The IRB overview assignment is a 20 minute video with a little quiz at the end. It's not overly complicated; a lot of ethics comes down to pretty basic common sense and being a decent person in dealing with other people.

You'll also get kind of the more in depth issues with the IRB and human subjects research with the CITI ethics training, which I think some of you may have already completed for other courses. If you've already completed this, you just need to submit your certificate for that assignment.

For the CITI training, you should complete either group 1, 2, or 3, depending on your research interest.

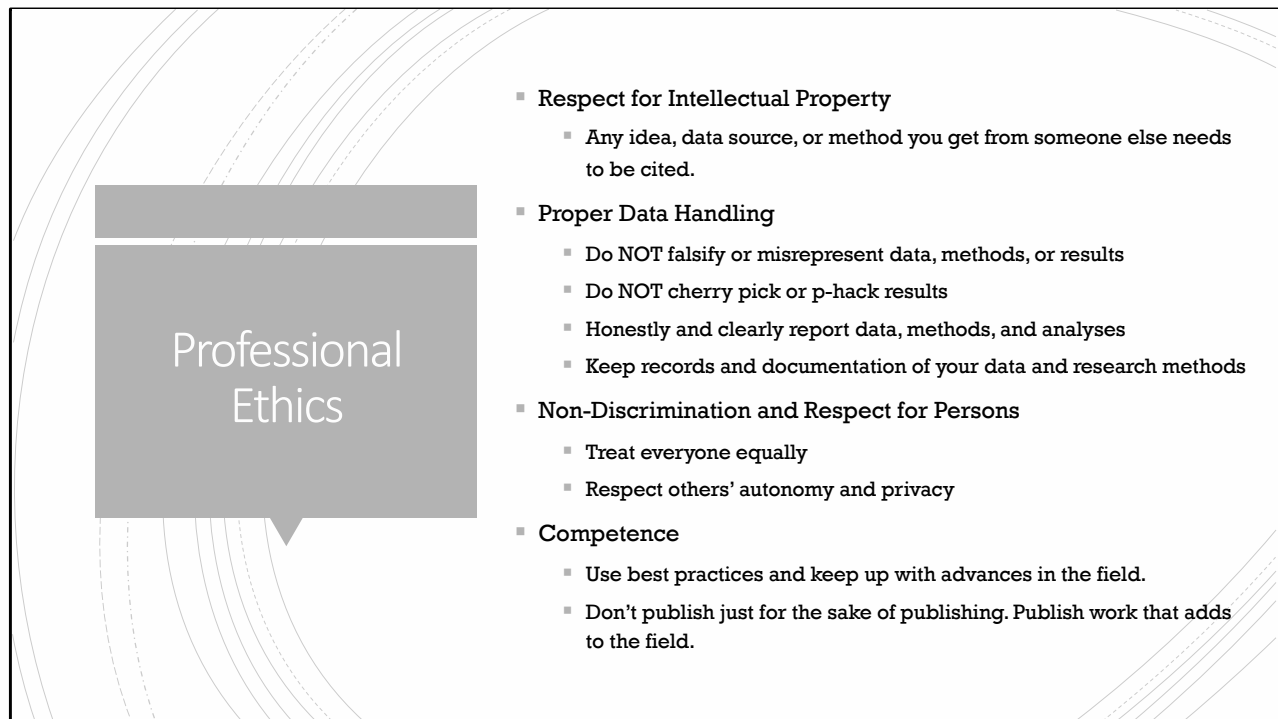
Group 1 is for if you are interested in the healthcare research that training covers the applications of collecting health data and how you deal with like HIPAA and some other kind of health specific things.

The kind of most applicable training for everyone is going to be Group 2, which is for social and behavioral research.

It covers any sort of human subjects research useful for both primary data collection methods like surveys or interviews but also for working with secondary data on individual like social media data.

So even if you're not, you know.

There's also a Group 3 which is just if you're working with data or specimens. If you're working with the type of data that maybe is only tangentially related to humans, like aggregated health data or aggregate other type of demographic data or even non human data where it might impact humans, but not necessarily about them such as many AI applications, Then you can take the Group 3 for data and lab specimens.



Moving on to the kind of next component of ethic, professional ethics and professional norms.

There's the respect for intellectual property which is at least somewhat regulated and like determined by different laws around copyright and trademarking and just general intellectual property.

This is relevant to academic integrity; if you're going to use an idea, data source, a method, basically anything where you're getting someone else's ideas to use in your own work, you need to cite those ideas. You need to give credits and respect any limits on that past work. Sometimes, you'll see that there's certain restrictions on open source software or data which may only be available for non-commercial uses or which you might need to seek permissions for and agree to terms of use.

In terms of being a good scientist, there's ethics around proper data handling that you don't falsify or misrepresent data, methods or results that you don't cherry pick your results to just get what you want, that you honestly and transparently report any data, methods, analysis, and basically anything you did. It doesn't matter if other people would necessarily agree that that was the right thing to do; the important thing is that you document it because as long as it's documented, then you can have an open, honest discussion about whether or

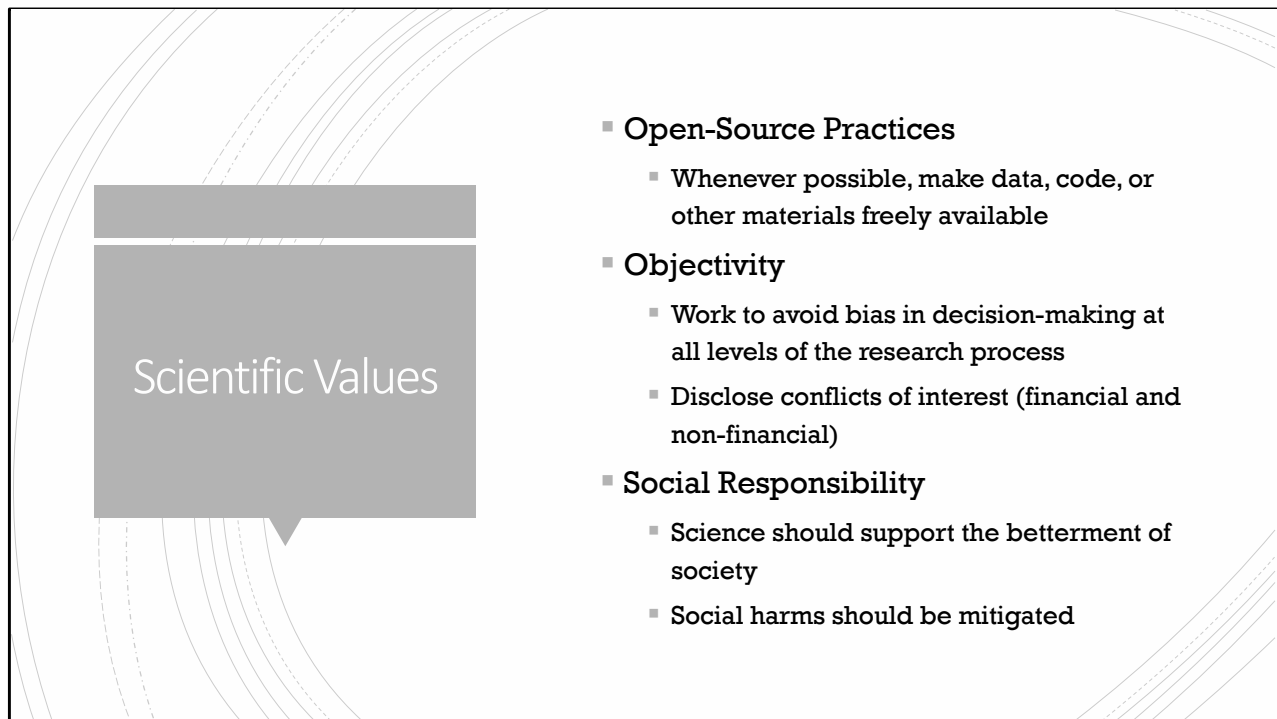
not that is the right analytic method or the right preprocessing method. But if it's not documented anywhere, then, there comes that issue of misrepresentation.

Going back kind of to some of the IRB principles, non-discrimination and respect for persons.

One ongoing debate around AI is the extent to which certain data should be considered private and off limits for the use in research or commercial tools when clear consent for its use was not given. Specific use cases will vary but important to keep in mind individual's autonomy and privacy.

Final issue is competence; as a researcher, you should do your best to keep up with advances and to use best practices.

You shouldn't just do things for the sake of doing things, but design research with clear, rational, logical justification for the questions you choose to answer for the data you choose to collect, etc.



There are also some scientific values and this is where coming from more of an academic space rather than the kind of more applied corporate or industry space.

creates some conflict because while in the academic space these are very strongly held values, they're not necessarily the most applicable if your goal is profit for corporate applications.

These out there because I do think they're important to at least consider and important if you are working on more academic research, but may not be the most applicable in non academic settings.

Things like open source practices, making your data, your code, your materials freely available whenever possible, as long as it's not potentially violating any data privacy or data confidentiality concerns .

Objectivity is another value. It's impossible to not be biased; we're all biased.

That's just a psychological fact of humanity, it seems.

But what's important is that you try to identify those biases and really consider how they're impacting your decision making in the research process and making sure they're not guiding your judgments in ways that might not be best practices, and then disclosing conflicts of interest, both financial and non financial.

Disclosing conflicts of interest is often a requirement for certain publication of

research. Depending on where you're publishing, and for some other kind of funding of research but it's just a good practice, if there's potential financial gains to be made from research turning out one way over the other; people have the right to know, especially in the healthcare domain, that you know that might not be the most objective research process.

And then finally social responsibility. We should try to be good people and make sure we're not harming humanity and the earth within our best judgment.

Importance of Data Science Ethics

- Models built by data scientists/analysts can have implications and impacts on peoples' lives for good or bad.
- While an emerging topic, little has been done to formalize ethics in data science (compared to the strict federal and international regulations governing human subjects research).
- Even well-intentioned models might cause unintended negative consequences if the analyst doesn't consider bias in the data or model.

Regardless of what you're working on, the models the algorithms, the systems you're building can, can and most likely do have implications and impacts on real people's lives, even if you're working on data or systems that don't necessarily have to do with humans.

Even something like supply chain still impact humans, because at the end of the day it's humans behind all of this and humans using all of this information.

So it's important to think about what, how, what you're doing.

A few years ago this really there was very little formalized in ethics around data science.

Even now, compared to the regulations around human subjects research, it's still not nearly as formalized and regulated, but it has become more so.

In the EU and other parts of the world, legislation like the GDPR and EU AI Act have put frameworks in place for oversight of the use of data

There's not a lot of regulations at a federal level in the US, but there are increasingly state level adopted legislation similar to the GDPR around data privacy and data use, so it's increasingly something that a lot of governments are regulating.

Even well-intentioned models can cause unintended negative consequences if you don't consider the implications, the impacts the assumptions you're making

about the data about the models, about the use of it. There's a lot that can go wrong.

Even if you have the best of intentions, you know a lot of the healthcare applications often have really good intentions of trying to help people increase access to health care or improved diagnosis of conditions.

But it's not a giant leap to take those same models, those same algorithms and apply them in a more, let's say, negative way, say in like health insurance, to deny people coverage based off of predicted probabilities of getting some illness.

Obviously that has very negative consequences on real people.

So really thinking through all of the implications of your work is important, even if you don't necessarily have the power to do anything about it.

Because at the end of the day, each individual's responsibility for these things is questionable but to the extent that you do have the power and the ability to respect these ethics and to consider the issues you should.